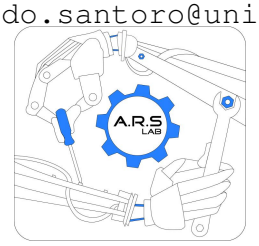
Elements of Optimal Control

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Advanced Robotics and Autonomous Systems

Notes

Optimal Control

Given a dynamic system:

$$\dot{x}(t) = f(x(t), u(t))$$

the theory of **optimal control** is based on finding a proper input function that "**makes optimal**" a certain objective function $J(x(t), u(t))$

While PID and classical techniques take care of typical step-response parameters (rise time, set-up time, overshot, etc.), optimal control let designers consider the **cost of control action**.

Notes

Feedback on the Complete State

Let us consider a dynamic system:

$$\dot{x}(t) = f(x(t), u(t))$$

whose state is completely observable, i.e. we can extract/estimate, as output, the overall **state vector** $x(t)$

Let us also consider a **target state** $\overline{x}$, i.e. a vector that specifies the final state we want to reach

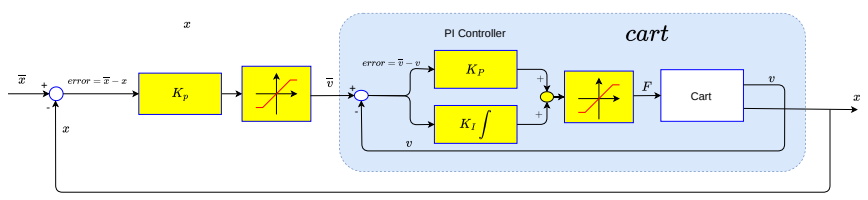
We want to **determine a controller** that is able to drive the system towards the target state

Notes

Example: Position Control of a Cart

In the problem of position control for a cart, we used two cascading controllers:

- Inner loop for speed
- Outer loop for position



In other words:

- we get the overall state $\mathbf{x} = \begin{bmatrix} v \\ p \end{bmatrix}$
- extract single variables and
- send them through different feedback loops

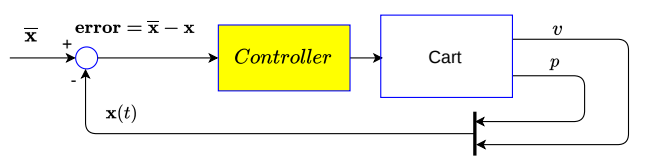
Notes

Example: Position Control of a Cart

But we can also think that reaching our target position $\overline{p}$ implies that

- $p = \overline{p}$
- $v = 0$

in order words, we consider out target as a **target state** $\overline{\mathbf{x}} = \begin{bmatrix} 0 \\ \overline{p} \end{bmatrix}$, put in feedback the overall state vector, and design the proper controller



In such a case:

- we have no more scalars but **vectors** (also the error is a vector)
- the controller receives an error vector and must be able to handle it

Notes

Back to Optimal Control

Hypotheses

- Let us consider a dynamic system:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t))$$

  whose state is completely observable, i.e. we can extract/estimate, as output, the overall **state vector** $\mathbf{x}(t)$

- Let us consider a **target state** $\bar{\mathbf{x}}$, i.e. a vector that specifies the final state we want to reach

- Let us consider a **cost function**:

$$L(\mathbf{x}, \mathbf{u})$$

  measuring the cost of a certain state **x** and of a certain input **u**

Notes

Back to Optimal Control

Optimal Control

Optimal Control means to find a proper law $\mathbf{u}(t)$ that

- Drives the system towards $\overline{\mathbf{x}}$ at time t_E
- **minimize the cost**:

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^{t_E} L(\mathbf{x}(t), \mathbf{u}(t))dt$$

Notes

Definitions

Matrices Positive-Definite

A matrix $A \in \mathcal{R}^{n \times n}$ is **positive-definite** if for each vector $\mathbf{x} \in \mathcal{R}^{n \times 1}$ we have

$$\mathbf{x}^T A \mathbf{x} > 0$$

If $eig(A) > 0$, then the matrix is **PD**

Matrices Positive-Semidefinite

A matrix $A \in \mathcal{R}^{n \times n}$ is **positive-semi-definite** if for each vector $\mathbf{x} \in \mathcal{R}^{n \times 1}$ we have

$$\mathbf{x}^T A \mathbf{x} \geq 0$$

If $eig(A) \geq 0$, then the matrix is **PSD**

Notes

Linear Quadratic Regulator (LQR Control)

Notes

LQR Control

LQR Control is a optimal control strategy that considers **linear systems**:

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

and uses a **quadratic form** of the objective function:

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^\infty (\mathbf{x}(t)^T Q\, \mathbf{x}(t) + \mathbf{u}(t)^T R\, \mathbf{u}(t))dt$$

where

- $Q \in \mathcal{R}^{n \times n}$ must be positive-semidefinite, it weights the cost of the state
- $R \in \mathcal{R}^{n \times n}$ must be positive-definite, it weights the cost of the input

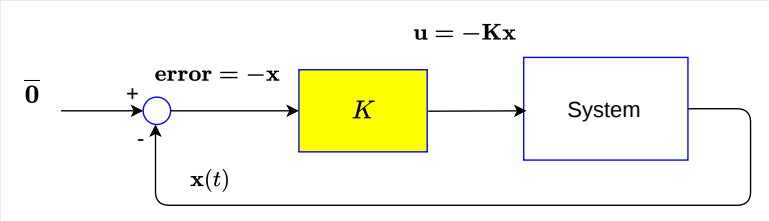
Notes

LQR Control

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^{\infty} (\mathbf{x}(t)^T Q\, \mathbf{x}(t) + \mathbf{u}(t)^T R\, \mathbf{u}(t))dt$$

The objective is to find a law $\mathbf{u}(t) = -K\mathbf{x}(t)$ that leads state to origin $\mathbf{x}(t) \to \mathbf{0}$ and minimize the objective function

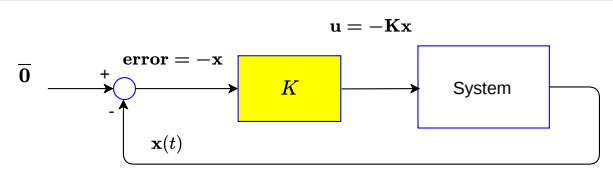


Notes

LQR Control: Why origin?

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^{\infty} (\mathbf{x}(t)^T Q\, \mathbf{x}(t) + \mathbf{u}(t)^T R\, \mathbf{u}(t))dt$$



The integral function must be summable, i.e. the integral **must exist**

If $\mathbf{x}(t) \to \mathbf{0}$, then also $\mathbf{u}(t) \to \mathbf{0}$, so the integral can be computed

Notes

LQR Control: Why “quadratic”?

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^\infty (\mathbf{x}(t)^T Q\, \mathbf{x}(t) + \mathbf{u}(t)^T R\, \mathbf{u}(t))dt$$

The forms $\mathbf{x}^T Q\, \mathbf{x}$ and $\mathbf{u}^T R\, \mathbf{u}$ are **scalar**, so the integrating function is scalar as well

If Q is diagonal $= diag\{q_1, \ldots, q_n\}$, $q_i \geq 0$, we have:

$$x^T Q\, x = x_1^2 q_1 + x_2^2 q_2 + \cdots + x_n^2 q_n$$

The same applies to $\mathbf{u}^T R\, \mathbf{u}$, so they are **quadratic forms**

Notes

LQR Control: Why “quadratic”?

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^{\infty} (\mathbf{x}(t)^T Q\, \mathbf{x}(t) + \mathbf{u}(t)^T R\, \mathbf{u}(t))dt$$

- The cost must be **positive**
- The integrating function must be **scalar**
- The quadratic form weights **stronger than linear** the distances from the origin

Notes

LQR Control: Meaning of the "cost"?

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^\infty (\mathbf{x}(t)^T Q\, \mathbf{x}(t) + \mathbf{u}(t)^T R\, \mathbf{u}(t))dt$$

Input Cost $\mathbf{u}^T R\, \mathbf{u}$

- The higher the input the higher the energy required
- An high cost R makes the control slower and smoother, in order to save energy

State Cost $\mathbf{x}^T Q\, \mathbf{x}$

- The higher the state the higher its distance from the origin
- An high the cost Q makes this distance "more important" thus producing a more aggressive control

Notes

LQR Control: Solving the problem

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

$$\min_{x,u} J(\mathbf{x}, \mathbf{u}) = \min_{x,u} \int_0^\infty (\mathbf{x}^T Q\, \mathbf{x} + \mathbf{u}^T R\, \mathbf{u}) dt$$

The (constant) **optimal gain** is:

$$K = R^{-1} B^T\, P$$

where $P \in \mathcal{R}^{n \times n}$ is the solution of the **Continuous-time Algebraic Riccati Equation (CARE)**

$$A^T\, P + P\, A - P\, B\, R^{-1}\, B^T\, P\, B + Q = 0$$

Notes

LQR Control: Solving the problem

$$\mathbf{x}(k+1) = A\,\mathbf{x}(k) + B\,\mathbf{u}(k)$$

$$\min_{x,u} J(\mathbf{x},\mathbf{u}) = \min_{x,u} \sum_{k=0}^{\infty} (\mathbf{x}(k)^T Q\,\mathbf{x}(k) + \mathbf{u}(k)^T R\,\mathbf{u}(k))dt$$

The (constant) **optimal gain** is:

$$K = (R + B^T\ P\ B)^{-1} B^T\ P\ A$$

where $P \in \mathcal{R}^{n \times n}$ is the solution of the **Discrete-time Algebraic Riccati Equation (DARE)**

$$P = Q + A^T\ P\ A - A^T\ P\ B\ (R + B^T\ P\ B)^{-1} B^T\ P\ A$$

Notes

Optimal Position Control in the Cart

Notes

The Cart: State Space Representation

$$\begin{bmatrix} \dot{v} \\ \dot{p} \end{bmatrix} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \end{bmatrix} + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

$$\dot{x} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

Let:

$$A = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} \qquad u = f$$

Notes

The Cart: State Space Representation

Let's consider $M = 1.0$ and $b = 0.9$:

$$\mathbf{x} = \begin{bmatrix} v \\ p \end{bmatrix} \qquad A = \begin{bmatrix} -0.9 & 0 \\ 1 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Let's discretize:

$$\widetilde{A} = I + A\Delta T \qquad \widetilde{B} = B\Delta T$$

Notes

LQR with target different than origin

Notes

LQR Control with target $\neq$ origin

Specifying a certain target

Let us consider we want to reach a target $\bar{\mathbf{x}}$ different than the origin and let's consider that out initial state is $\mathbf{x_0}$

We can apply a translation to the state and compute a new state vector $\mathbf{x_t}$:

$$\mathbf{x}^{(t)} = \mathbf{x} - \bar{\mathbf{x}}$$

The new system will thus start from state:

$$\mathbf{x}_0^{(t)} = \mathbf{x}_0 - \bar{\mathbf{x}}$$

and its target will be the:

$$\bar{\mathbf{x}}^{(t)} = \bar{\mathbf{x}} - \bar{\mathbf{x}} = \mathbf{0}$$

Notes

LQR Control with target $\neq$ origin

Specifying a certain target: Algorithm

Given a target $\bar{\mathbf{x}}$

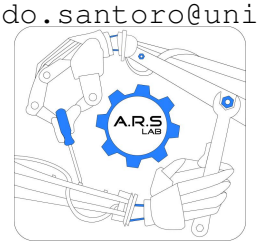
1. Compute the optimal gain
2. Set the state to your desired initial state $\mathbf{x_0}$ minus the target state: $\mathbf{x} = \mathbf{x_0} - \bar{\mathbf{x}}$
3. Let the system evolve using $\mathbf{x}$
4. Compute (for each step) the real state as $\mathbf{x} + \bar{\mathbf{x}}$

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